Cambridge (CIE) IGCSE Physics
Paper 6: Alternative to Practical (Set B)

Saturday 26 April 2025

Morning (Time: 1 hour 0 minutes)

Total marks

/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

1 (a) Students are investigating how the use of a lid or insulation affects the rate of cooling of hot water in a beaker. They use the apparatus shown in Fig. 2.1.

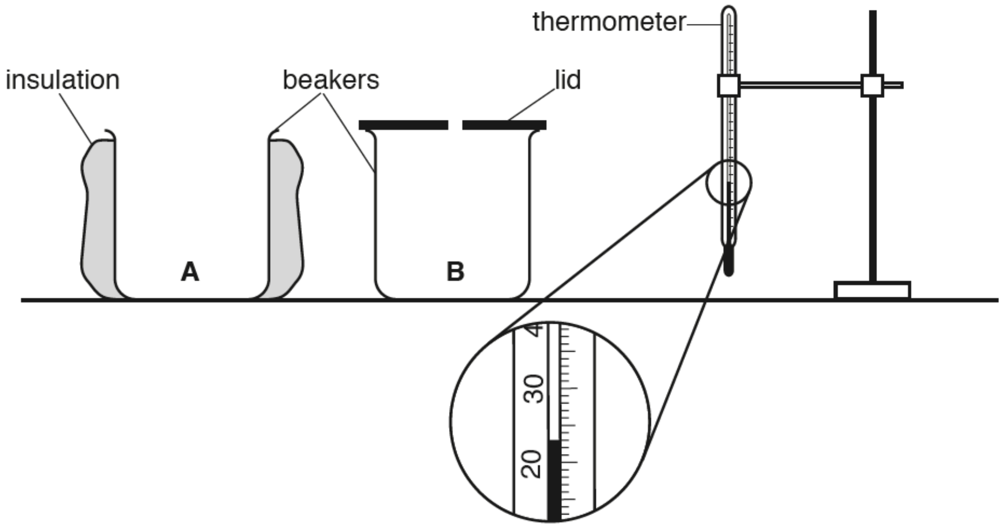

Fig. 2.1

Record the room temperature θ_R in °C shown on the thermometer in Fig. 2.1.

θ_R = ...

(1 mark)

(b)
- 100 cm^3 of hot water is poured into beaker **A** and the initial temperature θ is recorded in Table 2.1.
- The temperature θ of the water at times t = 30 s, 60 s, 90 s, 120 s, 150 s and 180 s are shown in Table 2.1.
- This process is repeated for beaker **B**.

Complete the headings and the time column in Table 2.1.

Table 2.1

	beaker **A** with insulation	beaker **B** with a lid
t/	*θ*/	*θ*/
0	83.0	86.0
	79.0	84.0
	75.5	82.5
	73.0	81.0
	71.0	80.0
	69.5	79.0
	68.5	78.5

(2 marks)

(c) Write a conclusion stating whether the insulation or the lid is more effective in reducing the cooling rate of the water in the beakers in this experiment.

Justify your answer by reference to the results.

(2 marks)

(d) One student thinks that the experiment does not show how effective insulation is on its own or how effective a lid is on its own.

Suggest an additional experiment which could be used to show how effective a lid or insulation is.

Explain how the additional results could be used.

....

....

(2 marks)

(e) (i) Calculate x_A, the average cooling rate for beaker **A** over the whole experiment. Use the readings for beaker **A** from Table 2.1 and the equation

$$x_A = \frac{\theta_0 - \theta_{180}}{T}$$

where T = 180 s and θ_0 and θ_{180} are the temperatures at time t = 0 and time t = 180 s. Include the unit for the cooling rate.

x_A = ... [2]

(ii) Students in another school are carrying out this experiment using identical equipment.

State why they should make the initial temperature of the water the same as in this experiment if they are to obtain average cooling rates that are the same as in Table 2.1. Assume that the room temperature is the same in each case.

Use the results from beaker **A** to explain why this factor should be controlled.

[2]

(4 marks)

2 (a) A student determines the resistances of some filament lamps.

Fig. 2.1 shows the first circuit she uses.

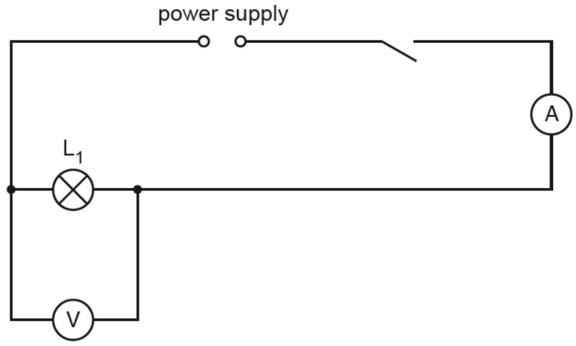

Fig. 2.1

(i) Record the potential difference V_1 across the lamp L_1, as shown on the voltmeter in Fig. 2.2.

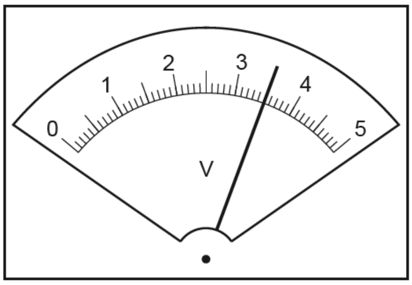

Fig. 2.2

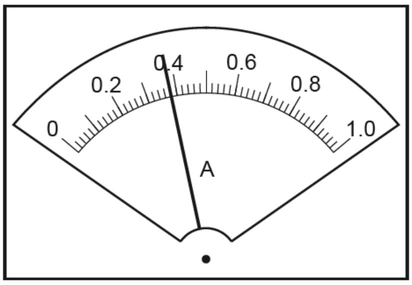

Fig. 2.3

V_1 = ... V [1]

(ii) Record the current I_1 in the circuit, as shown in Fig. 2.3.

I_1 = ... A [1]

(iii) Calculate the resistance R_1 of the filament of lamp L_1. Use the equation $R_1 = \frac{V_1}{I_1}$.

Include the unit.

R_1 = ... [2]

....

....

....

....

(4 marks)

(b) The student disconnects the voltmeter. She connects lamp L_2 in series with lamp L_1. She connects the voltmeter across lamp L_2.

She measures the current I_2 in the circuit and the potential difference V_2 across lamp L_2.

I_2 = ... 0.30 A ..... V_2 = ... 1.7 V .....

Calculate the resistance R_2 of the filament of lamp L_2. Use the equation $R_2 = \frac{V_2}{I_2}$.

R_2 = ...

The student disconnects the voltmeter. She connects lamp L_3 in series with lamps L_1 and L_2. She connects the voltmeter across lamp L_3.

She measures the current I_3 in the circuit and the potential difference V_3 across lamp L_3.

I_2 = ... 0.26 A ..... V_2 = ... 1.2 V .....

Calculate the resistance R_3 of the filament of lamp L_3. Use the equation $R_3 = \frac{V_3}{I_3}$.

R_3 = ...

....

(1 mark)

(c) Calculate $R_1 + R_2 + R_3$. Give your answer to a suitable number of significant figures for this experiment.

$R_1 + R_2 + R_3$ = ...

....

(1 mark)

(d) Some students make suggestions about the results of the experiment.

Suggestion **A**: $R_1 + R_2 + R_3$ should be equal to $3 \times R_1$

Suggestion **B**: $R_1 + R_2 + R_3$ should be less than $3 \times R_1$

Suggestion **C**: $R_1 + R_2 + R_3$ should be greater than $3 \times R_1$

State which suggestion **A**, **B** or **C** agrees with your results. Justify your answer by reference to your results.

....

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(2 marks)

(e) Draw a circuit diagram to show the circuit used in part (b) with all three lamps connected in series.

(3 marks)

3 (a) A student is determining the refractive index *n* of the material of a transparent block.

Fig. 3.1 shows the outline **ABCD** of the transparent block.

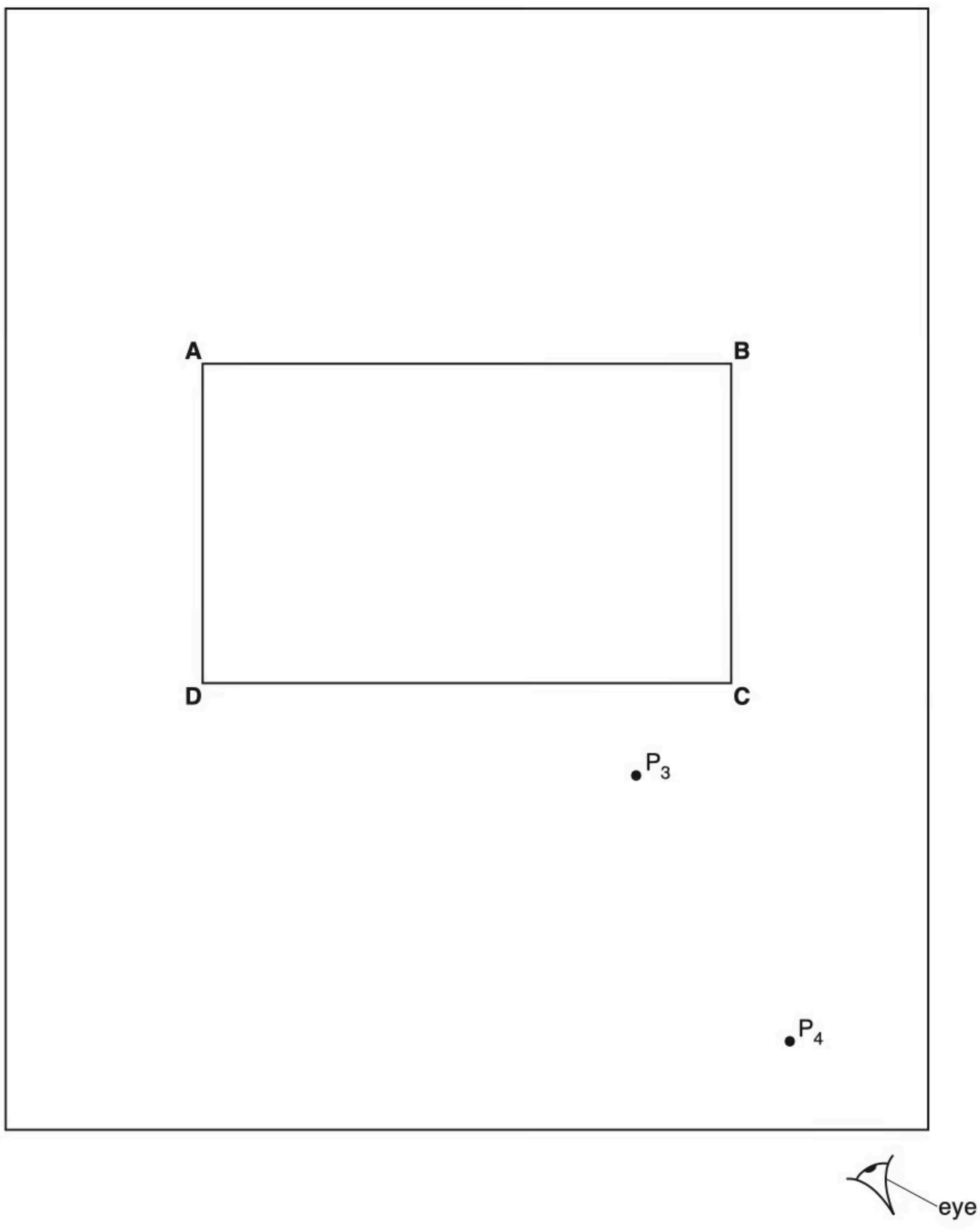

Fig. 3.1

(i) On Fig. 3.1:

- draw a normal **NL** at the centre of side **AB**
    - continue the normal so that it passes through side **CD** of the block
    - label the point **F** where **NL** crosses **AB**
    - label the point **G** where **NL** crosses **CD**.

[1]

(ii) Draw a line **EF** at an angle i = 30° to the left of the normal and above side **AB**.

[1]

(iii) Mark the positions of two pins P_1 and P_2 on line **EF** placed at a suitable distance apart for this type of ray-tracing experiment.

[1]

....

....

....

(3 marks)

(b) The student observes the images of P_1 and P_2 through side **CD** of the block so that the images of P_1 and P_2 appear one behind the other.

He places two pins P_3 and P_4 between his eye and the block so that P_3, P_4 and the images of P_1 and P_2 seen through the block, appear one behind the other.

The positions of P_3 and P_4 are marked on Fig. 3.1.

(i) Draw a line joining the positions of P_3 and P_4. Continue the line until it meets the normal **NL**.

Label the point **H** where the line meets side **CD**. Draw the line **FH**.

[1]

(ii) Measure and record the length *a* of the line **GH**.

a = ... [1]

(iii) Measure and record the length *b* of the line **FH**.

b = ... [1]

(iv) Calculate the refractive index *n* using the following equation:

$$n = \frac{0.5b}{a}.$$

n = ... [1]

(4 marks)

(c) The student repeats the procedure using the angle of incidence i = 45°.

a = 3.2 cm

b = 6.9 cm

Calculate the refractive index n, using the equation $n = \dfrac{0.71b}{a}$.

n = ...

....

(1 mark)

(d) The student expected the two values of refractive index n obtained in this experiment to be equal.

State **two** difficulties with this type of experiment that could explain any difference in the two values of n.

....

....

(2 marks)

(e) A student suggests precautions to take in this experiment to obtain reliable results. Tick **one** box to indicate the most sensible suggestion.

- ☐ Carry out the experiment in a darkened room.
- ☐ Use pins that are taller than the height of the block.
- ☐ View the bases of the pins.
- ☐ View the pins with one eye closed.

....

(1 mark)

4 A student is investigating the work required to pull a box containing some masses up a sloping wooden board. Fig. 4.1 shows the board and the box.

Plan an experiment to investigate how the work required to pull the box up the slope depends on the mass of the box and its contents.

Work done is calculated using the equation:

work done = force × distance moved in the direction of the force.

The following apparatus is available to the students:

a wooden board a box with a length of string attached a selection of masses that fit in the box a metre rule an electronic balance.

In your plan, you should:

- list any other apparatus that you would use
- explain briefly how you would carry out the investigation, including the measurements you would take
- state the key variables that you would control
- draw a suitable table, with column headings, to show how you would display your readings (you are **not** required to enter any readings in the table)
- explain how you would use the results to reach a conclusion.

You may add to the diagram if it helps your explanation.

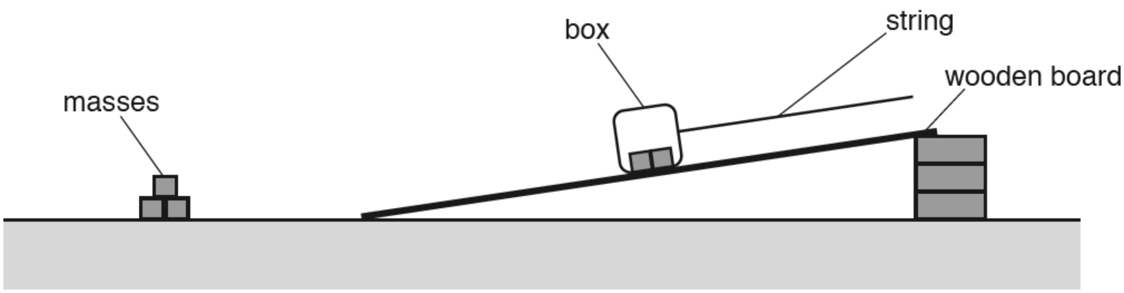

Fig. 4.1

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(7 marks)